Lesson 2.7 - Side by Side Stem and Leaf Plots

> **Video:** Side by Side Stem and Leaf Plots by The Organic Chemistry Tutor **Chapter:** 2 - Graphs and Tables

Idea

A **side-by-side** (back-to-back) stem-and-leaf plot compares **two groups** that share one stem column:

```
Group A leaves | Stem | Group B leaves
```

- **Right side:** leaves increase **left** → **right**, as usual.
- **Left side:** leaves increase **right** → **left**, going *away* from the stem. Read the left side backwards: `8 | 1` means **18**.

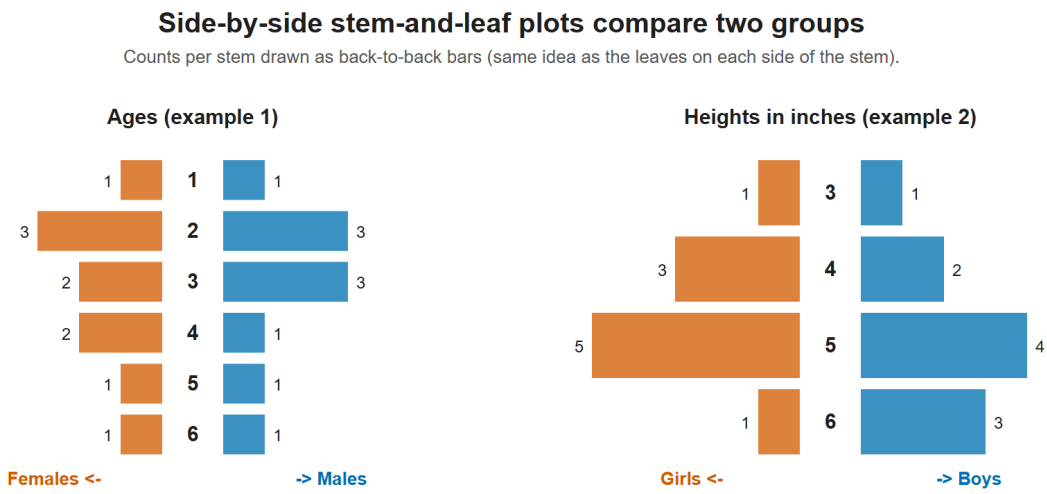
Example 1 - ages of females and males

Females (leaves)	Stem	Males (leaves)
8	1	9
8 7 5	2	2 6 9
5 2	3	5 5 6
7 3	4	6
2	5	8
5	6	3

Key: `8 | 1 |` = 18 (female); `| 1 | 9` = 19 (male).

- Males: 19, 22, 26, 29, 35, 35, 36, 46, 58, 63
- Females: 18, 25, 27, 28, 32, 35, 43, 47, 52, 65

On the female side the smallest leaf sits **next to** the stem. For the 20s, the leaves read `8 7 5`, which means 25, 27, 28.



The number of leaves on each side of each stem, drawn as back-to-back bars. The heights example makes the comparison easy to see: the girls cluster in the 50s while the boys spread further into the 60s.

Example 2 - heights (inches) of girls and boys

Girls (leaves)	Stem	Boys (leaves)
9	3	7
8 6 5	4	7 8
6 4 3 2 1	5	1 6 7 8
0	6	2 3 5

- Girls: 39, 45, 46, 48, 51, 52, 53, 54, 56, 60
- Boys: 37, 47, 48, 51, 56, 57, 58, 62, 63, 65

Comparing the two sides shows that more boys are in the 60s, while the girls are concentrated in the 50s.

Key takeaways

- One shared stem column, with one group's leaves on each side.
- Leaves on the **left** are written in reverse, smallest next to the stem.
- Side-by-side plots make it easy to compare the **center** and **spread** of two groups.